THOMAS MARIANO, E.I.T. | SYSTEMS ENGINEER | SECRET CLEARANCE

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SKILLS

Languages & Data: Python | JavaScript | SQL | Java | C++ | MATLAB | HTML | CSS | JSON
Software & Systems: Git/GitHub | Docker | Kubernetes | Linux | Windows Server | PowerShell | PyQt6 | Automated Testing
Solutions Delivery: Requirements Gathering | Technical Demos | User Training | Stakeholder Communication
Additional Tools: GNU Radio | SDR | Raspberry Pi | SolidWorks | Fusion 360

WORK EXPERIENCE

Systems Engineer, Senior Associate, NASA - *GSFC - Peraton (contractor)* **October 2024 - Present**
Supporting Human Space Flight Missions as a Search and Rescue Engineer

- Built a Python/PyQt6 MEOSAR scheduler that automates satellite visibility calculations and antenna tracking assignments, reducing mission planning from 2 days to less than a minute
- Gathered requirements from internal and external users, delivered demos and training, and distributed versioned application bundles and Python source through internal Git, incorporating feedback into subsequent releases
- Developed custom monitoring tools using Python and Docker to query and centralize health, maintenance, and update status across 180+ systems, helping operators identify faults and prioritize maintenance
- Analyzed 70+ launch and landing events using Python to identify coverage gaps and beacon-data overload; recommendations were implemented to filter hundreds of interfering data records
- Developed an integrated Python/JavaScript simulator that transforms exported tracking schedules into interactive satellite playback and geometric coverage analysis, with video export to support mission planning and review
- Integrate and troubleshoot mission software, Linux/Windows servers, networks, and RF interfaces across 180+ systems, using telemetry and logs to diagnose faults and validate corrective actions
- Integrated a 24-signal phased-array antenna by defining requirements, configuring software and networking, executing acceptance tests, and resolving integration issues in coordination with vendors and operators
- Developed and successfully tested a remotely controlled recovery-beacon prototype using Python, GNU Radio, and Raspberry Pi, integrating GPS, Iridium communications, and SDR-based transmission controls
- Built a Python tool to structure human spaceflight preparation checklists and retain completion history, supporting consistent readiness reviews and emergency-response preparation

TECHNICAL PROJECTS

MEOSAR Scheduling & Simulation Platform | Project Link

- Implemented scheduling algorithms using Skyfield visibility calculations, bounded beam search, and antenna-satellite matching to account for frequency bands, antenna availability, and contact retention
- Refactored scheduling and configuration workflows into responsibility-focused modules, with atomic JSON configuration writes, validated schedule editing, and machine-readable exports
- Integrated a Python orbital-data backend with a JavaScript globe through PyQt6-WebEngine and JSON, enabling schedule-driven playback, per-location geometric coverage analysis, and scheduler-simulator disagreement diagnostics
- Built asynchronous video export using timestamp-specific PNG frames, QProcess, and FFmpeg, with progress reporting, cancellation, and restoration of application state

GNSS-to-TLE Generation & Validation Pipeline

- Developed and integrated a Python pipeline that generates SGP4-compatible TLEs from GNSS navigation data for all MEOSAR satellites used by the system, supplying an internal orbital-data catalog to MEOLUT/MCC software
- Evaluated generated orbits against independent SP3 reference data for four GPS satellites, measuring 22–49 m RMS position differences over initial two-hour arcs and assessing error growth over longer forecasts
- Added 18 automated tests covering parsing, satellite-health checks, time and coordinate transformations, orbital fitting, and validation calculations; evaluated chronologically withheld trajectory samples

EDUCATION

Georgia Institute of Technology - M.S. in Computer Science **Expected May 2028**
University of Maryland, Clark School of Engineering - B.S. in Mechanical Engineering **May 2024**

CERTIFICATIONS

FE Mechanical (E.I.T.) - Passed **May 2025**

AWARDS

Robert H. Goddard Team Award - NASA, Supporting the Artemis II mission **June 2026**